



HEMPOXIES™ FORMULA 8: THE INDUSTRIAL BRIDGE

RESEARCH STRATEGY & LABORATORY COMMISSIONING PROTOCOL (TRL 3)

ORIGINATING ENTITY: [MARIELANDRYSPYSHOP.COM](https://marielandryspyshop.com) / HEMPOXIES MATERIALS DIVISION

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1.0 EXECUTIVE SUMMARY & MISSION

1.1 Abstract: Bridging the “Valley of Death”

The global composites industry faces an urgent need to decouple high-performance structural materials from the petrochemical supply chain while maintaining the mechanical rigor required for aerospace and automotive applications. While the theoretical **Formula 9 Hypothesis** proposes an ultimate “Structural Bullion” utilizing diamond nanothreads synthesized at 20 GPa, this technology currently resides at Technology Readiness Level (TRL) 1. **Hempoxies™ Formula 8** serves as the “Industrial Bridge,” a pragmatic, scalable formulation designed to validate the **Bio-Vitrimer Hypothesis** using existing atmospheric manufacturing infrastructure.

1.2 The TRL 3 Pivot (Bench Validation Strategy)

Previous iterations of this protocol (V5.0) prematurely targeted pilot scaling (20kg batches), which presented unacceptable exotherm risks. This finalized protocol restricts operations to **500g – 1kg** batch sizes to ensure personnel safety and chemical purity. The primary objective is to produce standard **ASTM Test Coupons** (150mm x 150mm) to physically validate three core performance metrics:

1. **Vitrimer Dynamics:** Can the material heal micro-cracks and be reprocessed via dynamic disulfide exchange at 140°C? [2, 19]
2. **Conductivity:** Does the “Mitlinitite” nanostructure achieve the percolation threshold required for Electromagnetic Interference (EMI) shielding (>40dB)? [15, 16]
3. **Mechanical Rigor:** Does the tensile strength exceed 300 MPa? [5]

1.3 Scientific Rationale: The Post-Predatory Philosophy

Formula 8 rejects the linear “take-make-waste” model of the petrochemical era. Instead, it asserts that materials must function as regenerative assets. This is achieved through the engineering of a **Covalent Adaptable Network (CAN)**. Unlike static epoxies, Formula 8 utilizes **Cystamine-based disulfide bridges**. Below the Topology Freezing Transition Temperature (T_v), the material behaves as a rigid, solvent-resistant thermoset. Above T_v (approx. 120-140°C), rapid bond exchange reactions allow the material to flow like a viscoelastic fluid, enabling repair and recycling without depolymerization [7, 21].

2.0 THE FORMULA 8 “INDUSTRIAL STACK” (TECHNICAL DEFINITION)

2.1 Bill of Materials (BoM)

All components are derived exclusively from *Cannabis sativa* L.

ID	Component Name	Source	Functional Role & Justification	Classification

C1	EHSO (High-Linoleic)	Seed	Matrix Monomer. Epoxidized Hemp Seed Oil with high oxirane content (>6.8%) via Prilezhaev epoxidation ensures rigid cross-linking density.	Standard
C2	Cystamine (Dimer)	Seed	Dynamic Hardener. Unlike simple cysteamine, the dimer form introduces the critical disulfide ($S - S$) bond directly into the crosslink.	NOVEL
C3	FGE	Hurd	Reactive Diluent. Furfuryl Glycidyl Ether. The furan ring provides high char yield and thermal stability while reducing viscosity.	NOVEL
C4	QF-MHL (Lignin)	Stalk	Rigid Co-Hardener. Quadruple-Function Mannich-Modified Lignin acts as a rigid aromatic scaffold and antioxidant.	NOVEL
C5	"Mitlinite" (Activated)	Bast	Nano-Reinforcement. KOH-Activated Carbon Nanosheets. Surface area $>2000\text{ m}^2/\text{g}$ for EMI shielding.	NOVEL
C6	Hemp Biochar	Waste	Micro-Filler. Pyrolyzed at 600°C. Increases bulk modulus and reduces resin cost. Carbon sink.	Standard
C7	Hemp Carbon Fibers	Bast	Macro-Reinforcement. Graphitized at 1500°C to achieve turbostratic carbon structure for tensile loads.	Standard

C8	Purified Bio-Silica	Ash	Coupling Agent. Dialysis-purified (Salt-Free) silica sol used to silanize fillers and prevent phase separation.	NOVEL
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2.2 Mathematical Framework (Stoichiometry)

To prevent plasticization, the total number of Active Hydrogens (H_{total}) from both Lignin and Cystamine must equal the number of Epoxide groups (E_{total}). The Cystamine load is specifically reduced to account for the contribution of the Lignin co-hardener.

The Hybrid Identity Equation:

$$Mass_{C2} = (E_{total} - H_{lignin}) \times AHEW_{C2}$$

- **Epoxide Equivalent Weight (EEW):** $\approx 245g/eq$ (Based on average molecular weight of EHSO).
- **Amine Hydrogen Equivalent Weight (AHEW):** $56.25g/eq$ (Based on Cystamine Dimer).
- **Optimized Formulation Ratio:**
 - **Resin Phase:** 100 parts (EHSO + FGE)
 - **Lignin Phase:** 15 parts (Provides ~15% of total crosslink density)
 - **Cystamine Phase: 18 parts** (Calculated to balance remaining epoxide groups).

3.0 STANDARD OPERATING PROCEDURES (THE 9 CORE SOPs)

SOP-01: Synthesis of Epoxidized Hemp Seed Oil (EHSO)

Objective: Convert double bonds to oxirane rings safely.

- **Scale:** 500g (Bench).
- **Safety: EXOTHERM RISK.** Thermocouple feedback loop required.

- **Process:**

1. Load 500g Hemp Oil, 77g Formic Acid, and 0.5g Sulfuric Acid (catalyst) into a 1L jacketed reactor.
2. Chill system to 15°C.
3. **Critical Step:** Add 30% Hydrogen Peroxide (H_2O_2) dropwise over 2 hours. Strictly monitor temperature; **abort if T > 55°C** to prevent explosive ring opening [23, 24].
4. Digest at 50°C for 6 hours.
5. **Quench:** Wash with Sodium Bicarbonate ($NaHCO_3$) to pH 7, then wash with Brine. Dry over Magnesium Sulfate ($MgSO_4$).

SOP-02: Synthesis of Mannich-Modified Lignin (QF-MHL)

Objective: Aminate lignin to increase reactivity.

- **Process:**

1. Dissolve Organosolv Lignin in Ethanol/Water (70/30) at 60°C.
2. Add Hemp-Derived Amine (Diethylene Triamine equivalent).
3. Add Formaldehyde dropwise to initiate the Mannich Reaction. Reflux at 80°C for 4 hours [17, 30].
4. Precipitate in acidic water (pH 2). Filter and vacuum dry.

SOP-03: Synthesis of Furfuryl Glycidyl Ether (FGE)

Objective: Create reactive diluent from hurd waste.

- **Process:**

1. Hydrolyze Hurd Pentosan to Furfural, then hydrogenate to Furfuryl Alcohol.
2. React with Bio-Epichlorohydrin and Sodium Hydroxide (NaOH) via Phase Transfer Catalysis [9, 11].
3. Vacuum distil to purify.

SOP-04: Synthesis of Cystamine Linker

Objective: Create the “Disulfide Bridge” for vitrimer dynamics.

- **Process:**

1. Oxidize Hemp-derived Cysteamine with Hydrogen Peroxide at 0°C to form Cystamine Dihydrochloride [32].
2. Neutralize with Sodium Hydroxide to release the free base diamine.
3. Extract into Ethyl Acetate and recrystallize to purity.

SOP-05: Synthesis of “Mitlinite” Nanosheets (Activated)

Objective: Create high-surface-area conductive carbon ($>2000\text{ m}^2/\text{g}$).

- **Safety: CORROSIVE HAZARD.** Molten KOH destroys glass and steel. **Nickel or Inconel Crucible Mandatory.**

- **Process:**

1. **HTC Phase:** Hydrothermal Carbonization of bast fiber at 180°C for 12 hours yields Hydrochar Spheres [14].
2. **Impregnation:** Mix Hydrochar with Potassium Hydroxide (KOH) at a 1:4 ratio [15].
3. **Activation:** Place in Nickel Crucible. Heat to **800°C** under Nitrogen flow for 1 hour. (Note: Spheres “pop” into nanosheets; Potassium intercalates graphite layers).
4. **Purification:** Cool under Nitrogen. Wash with dilute HCl (Caution: Hydrogen evolution), then DI water until pH 7.

SOP-06: Synthesis of Hemp Biochar

Objective: Create compressive micro-filler.

- **Process:**

1. Pyrolyze waste hurd at 600°C under Argon.
2. Ball mill to D50 < 50 microns [26].
3. Functionalize with C8 (Bio-Silica) solution.

SOP-07: Synthesis of Hemp Carbon Fibers

Objective: Create tensile backbone.

- **Process:**

1. **Stabilization (Critical):** Oxidative ramp in Air from 200°C to 280°C at **0.5°C/min**. Hold for 2 hours. This prevents the fibers from melting during carbonization [27].
2. **Carbonization:** Heat to 1000°C in Nitrogen.
3. **Graphitization:** Flash heat to 1500°C in Argon to align turbostratic carbon structure.

SOP-08: Synthesis of Bio-Silica Sol (Salt-Free)

Objective: Create compatibility agent.

- **Process:**

1. Extract Hemp Ash with NaOH to form Sodium Silicate [28].
2. Neutralize with HCl to form Silica Sol + Sodium Chloride (NaCl).
3. **Purification (Critical):** Dialyze the sol against flowing DI water (12kDa membrane) for 24 hours to remove NaCl [28, 29].
4. **Check:** Verify conductivity is $< 10 \mu S/cm$ before use.

SOP-09: Master Assembly & Curing (Coupon Fabrication)

Objective: TRL 3 Sample Production.

- **Process:**

1. **Phase A:** Disperse C5 (Mitlinite) and C6 (Biochar) in C3 (FGE) using ultrasonication.
2. **Phase B:** Mix Phase A into C1 (EHSO) and C4 (Lignin).
3. **Phase C:** Add C2 (Cystamine). Degas in vacuum planetary mixer (-0.1 MPa).
4. **Cure Profile:**
 - 60°C for 4 hours (Gelation).

- 100°C for 2 hours (Crosslinking).
- 140°C for 2 hours (**Vitrification Activation**: Dynamic exchange begins) [19, 21].

5. **Cooling**: Slow ramp down at 1°C/min to prevent internal stress.

4.0 VALIDATION & QUALITY ASSURANCE

Metric	Standard	Target	Rationale/Ref
Tensile Strength	ASTM D3039	> 300 MPa	Structural viability threshold [5].
Flexural Modulus	ASTM D790	> 40 GPa	Rigidity for automotive panels [36].
EMI Shielding	ASTM D4935	> 40 dB	Validates “Mitlinite” percolation network [6, 16].
Glass Transition (T_g)	ASTM E1356	> 80°C	Service temperature limit [37].
Vitrimer Exchange	Stress Relaxation	100%	Critical Proof of Concept [7, 19].

Failure Mode & Effects Analysis (FMEA):

- **Risk**: Salt contamination in Silica causing osmotic blistering. **Mitigation**: SOP-08 Dialysis + Conductivity Check.
- **Risk**: Crucible Failure during Activation. **Mitigation**: Use Nickel/Inconel hardware only. No Quartz/Glass.
- **Risk**: Fiber degradation/melting. **Mitigation**: Strict 0.5°C/min stabilization ramp in SOP-07.

5.0 APPENDIX 1: DESIGN OF EXPERIMENTS (The “Golden 10”)

Reduced from 50 to 10 high-value runs for TRL 3 validation.

Run	Code	Resin (g)	Lignin (g)	Cystamine (g)	Nano (g)	Hypothesis/Target
1	F8-BALANCED	100	15	18	0	Ideal Stoichiometry Baseline
2	F8-COND-1	100	15	18	0.5	Percolation Threshold Check
3	F8-COND-2	100	15	18	1.0	EMI Max Saturation
4	F8-STIFF	100	25	14	0.5	Max Lignin Load (High Stiffness)
5	F8-VIT-LO	100	15	18	0.5	Under-cured Flow Dynamics
6	F8-VIT-HI	100	15	18	0.5	Over-cured Brittle Limit
7	F8-RECY	100	20	16	0.25	Reprocessability Test
8	F8-ISO	100	15	18	0.25	Low-Energy Cure Profile
9	F8-FAIL	100	0	23	0	Control: No Lignin (Non-Stoichiometric)
						Final Optimized

6.0 APPENDIX 2: FUTURE ADDITIVE EXPLORATION (TRL 4+)

Objective: Expand the platform using “Pollution-as-Resource” once TRL 3 chemistry is stable.

1. **Microplastics (Upcycling):** Utilizing lignin (C4) as a surfactant to compatibilize PET/PP waste into the matrix [39].
2. **Terpenes (Limonene):** Use as a reactive diluent to adjust viscosity (requires Thiol-Ene adjustment) [42].
3. **Cannabinoids (CBD):** *Caution:* While CBD is a potent antioxidant, it may act as a plasticizer, lowering T_g . Must be isolated and tested for impact on modulus [44].
4. **Bio-Foaming:** Trapping metabolic CO_2 during cure to create structural insulation foam [40].

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AUTHORIZED FOR IMMEDIATE EXECUTION

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